MASTER GAME DESIGN DOCUMENT & COMPLETE ITEM COMPENDIUM // V2.0

SUB-LEASE

Physics-Driven Co-Op & Solo VR Liminal Furniture Heist & Vertical Extraction Sim

TARGET DEMOGRAPHIC	CORE INSPIRATION	ENGINE & PHYSICS	TARGET PRICE
Ages 15–30 (PCVR & Quest 3)	Lethal Company × R.E.P.O. × Boneworks	Unity 6 + PhysX 5 + OpenXR	$19.99 – $24.99

THE 10-SECOND HOOK: You and your crew breach a vertical maintenance hatch into the Backrooms for one reason: **millions of dollars of untaxed, pristine Mid-Century Modern furniture, crystal chandeliers, arcade cabinets, and anomalous sculptures sitting down there for free.** Rappel down vertical tubes into deeper liminal floors, steal heavy furniture, protect it from physics-driven monsters, and engineer pulleys, counterweights, and duct-tape contraptions to hoist a $3,500 haul back up a 100-foot shaft.

1. Your Loot IS Your Barricade

Every piece of furniture is both **Money ($)** and a **Physics Tool**. Wedging an $850 Mahogany Chair under a doorknob stops a monster—but every dent lowers its resale value!

2. Vertical Rappel Engineering

Sliding down a 100ft tube is easy. Getting a 220kg Marble Statue *back up* requires rigging 4x pulleys, dropping heavy counterweights, or inflating CO2 salvage bags.

3. 100% Universal Physics

Zero scripted lock-and-key puzzles. Flashlights, grappling hooks, ropes, and tape are physical rigidbodies that combine freely into traps, decoys, and vehicles.

1. THE 6-STAGE CORE GAMEPLAY LOOP (15–25 MINUTE RUNS)

1. Drop Hatch (0m)
Check 3-Day Quota, buy ropes/tape/grapples, anchor primary lines.

2. Rappel Tubes
Clip Waist Carabiner & descend vertical shafts (-30m to -600m).

3. Recon & Scan
Appraise furniture ($/kg), scan for Mimics with UV light & scales.

4. Prep & Rig
Tape caster wheels to sofas, wrap fragile loot, set tripwires.

5. Vertical Haul
Pulley-hoist or counterweight-lift loot up shafts under attack.

6. Cargo Scale
Weigh intact furniture & taped monsters, cash out, upgrade gear.

2. COMPLETE EQUIPMENT & ITEM COMPENDIUM (36 PHYSICAL TOOLS)

Every item clips physically to the player's **Utility Belt** or stows in the **Equipment Crate**, and obeys universal PhysX mass, joint tension, friction, and acoustic impulse rules.

Category A: Ropes, Tethers & Rappelling Gear

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
A1	**Hemp Twine Spool**	Starter	`Len: 30m | Break: 800N`	Basic door tying, shin-height tripwires, and light rappelling. Snaps if hoisting loads >80kg without a pulley or if chewed by *The Seep*.
A2	**Dynamic Climbing Rope**	$250	`Len: 80m | Break: 2,200N`	8% elastic stretch absorbs sudden kinetic shock when rappelling deep shafts or catching falling furniture mid-air.

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
A3	**Braided Steel Cable**	$650	`Len: 150m | Break: 6,000N`	Unbreakable by monsters, immune to sharp concrete edges, and **conducts electricity** between wall junction boxes and metal doors.
A4	**Industrial Bungee Cord**	$300	`Len: 15m | Spring k: 450N/m`	Stores elastic potential energy ($F = -kx$). Pull a heavy door back against a bungee, hold with thin twine, and let a monster trip it to slam at 60 mph.
A5	**Waist Brake Carabiner**	Starter	`Trigger Friction | Lock-Off`	Squeeze trigger to control rappel descent speed, or toggle Lock-Off to **hang mid-air inside a vertical shaft with both hands free!**
A6	**Motorized Belt Ascender**	$1,100	`Lift: 3,000N | Noise: 85 dB`	Battery winch that reels you and 120kg of furniture straight up a vertical shaft at 4 m/s—*but its loud whine draws The Tuning Fork!*

Category B: Grappling Hooks, Harpoons & Anchor Hardware

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
B1	**3-Pronged Throwing Grapple**	$200	`Mass: 3.5kg | Hold: 2,000N`	Swing by hand in VR like a pendulum and throw! Clamps onto pipes, vents, chandeliers, or **distant furniture** to "fish" loot across pools or up shafts.
B2	**Pneumatic Piton Harpoon**	$750	`Range: 35m | Hold: 5,000N`	Compressed-air gun that fires a tethered steel spike into solid concrete ceilings or **directly into a monster**—harpoon *The Ram* to a 300kg pillar!
B3	**Silent Magnetic Grapple**	$450	`Hold: 2,500N | Noise: 5 dB`	Rubber-coated neodymium magnet head. Locks onto metal air ducts, steel doors, and filing cabinets with **zero metallic clatter**.
B4	**Suction Cup Pitons (3-Pack)**	$220	`Hold: 1,500N | Reusable`	Stick onto any flat wall, glass window, or tile ceiling to create a custom rope anchor, pulley mount, or tripwire post where no pipe exists.
B5	**Compound Pulley Block**	$350	`Advantage: 2x or 4x Force`	Clip to a shaft top and thread rope through sheaves. Divides effective furniture weight by 4, letting one player hoist a 220kg marble statue by hand!
B6	**Ratchet One-Way Cleat**	$180	`One-Way Lock | Max: 4,000N`	Automatically locks hoisted rope progress so if you let go mid-haul to fight an entity, your sofa doesn't plummet back down the shaft.

Category C: Flashlights, Illumination & Optical Scanners

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
C1	**Angle-Head Clip Light**	Starter	`Bat: 15m | Click: 35 dB`	Clips to belt for hands-free rappelling, or **duct-tapes to a rolling chair / ceiling rope** to create a swinging pendulum spotlight decoy.
C2	**High-Lumen Halogen Flood**	$400	`Mass: 6kg | Beam: 10,000 lm`	**Shining its high beam directly into *The Counterweight*'s ceiling hole or at *The Seep* burns/blinds them**, forcing them to drop stolen loot or teammates!
C3	**UV Blacklight Inspector**	$300	`365nm UV | Bat: 25m`	Reveals invisible **hairline cracks** on fragile loot, highlights **fraying rope fibers** under tension, and exposes **glowing veins on sleeping Mimics!**
C4	**Hand-Crank Dynamo Lantern**	$180	`Infinite Bat | Crank: 60 dB`	Hang from a rope mid-room or tie to your loot cart for 360° light. Never runs out of power, but cranking it generates mechanical sound.
C5	**Strobe & Siren Flare**	$150	`30s Burn | 115 dB + Strobe`	Pull pin and throw (or tape onto *The Ram*!). Blinds light-sensitive monsters and deafens *The Tuning Fork* so you can extract loudly.
C6	**Laser Trip-Beam Pointer**	$140	`Range: 40m | Silent Alert`	Tape across a hallway or shaft opening; vibrates your belt receiver the instant a stalking entity breaks the laser line.

Category D: Tapes, Adhesives & Protective Packaging

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
D1	**Standard Grey Duct Tape**	$60	`20 Strips | 1,200N / strip`	Welds any two props into a single compound rigidbody; seals ceiling vents; locks door seams. Strips stack linearly in break strength.
D2	**Kevlar Strapping Tape**	$280	`15 Strips | 4,500N / strip`	Fiberglass-reinforced tape. Strong enough to **tape a sleeping Mimic's jaws shut** (4 strips) so you can haul the live monster up and sell it!
D3	**Acoustic Foam Tape**	$200	`12 Strips | Sound: -90%`	Wrap around furniture corners, rattling door handles, grapple claws, or boots to haul heavy loot past *The Tuning Fork* in dead silence.
D4	**Rubber Electrical Tape**	$150	`15 Strips | 100% Insulation`	Wrap around metal crowbars, steel cables, or electrified door handles so you can safely touch high-voltage traps that stun entities.
D5	**Heavy Bubble-Wrap Roll**	$180	`4 Uses | Dent Damage: -85%`	Wrap around fragile high-dollar loot (Chandeliers, Glass Tables, Arcade Cabinets) so shaft-wall impacts during hoisting don't ruin resale value.
D6	**Industrial Flypaper Strip**	$130	`4m Roll | Shear Drag: 900N`	Unroll across floors or inside vertical vents. Slows *The Seep* to a crawl and gums up charging enemies' limbs.

Category E: Moving-Company Hardware, Leverage & Surface Tools

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
E1	**Swivel Caster Wheels (4x)**	$140	`Friction: 0.015 | Max: 300kg`	**Duct-tape onto the bottom of any heavy sofa, safe, or wardrobe** to turn a 150kg object into a smooth rolling cart or downhill battering ram!
E2	**Steel Crowbar / Lever Pipe**	$120	`Len: 1.2m | Leverage: 5x`	Pry jammed doors, lift 300kg slabs over a fulcrum brick, hook over taut steel cables to **zip-line across atriums**, or crossbar double doors.
E3	**Inflatable CO2 Lift Bag**	$380	`Upward Lift: +95kg Buoyancy`	Tape to heavy furniture at the bottom of a shaft or flooded pool and pull the lanyard—inflates a salvage balloon that floats 95kg to the ceiling!
E4	**Slip-Detergent Jug**	$110	`4 Uses | Friction: 0.005`	Pour in front of open shafts or stairs so charging monsters slide off ledges; also lets you slide heavy crates across carpet with one hand.
E5	**Spring-Loaded Box Cutter**	$80	`0.1s Instant Sever`	Instantly slice taut ropes to drop suspended counterweights on enemies, or emergency-cut a snagged hoist line.
E6	**Stair-Climbing Dolly**	$320	`Tri-Wheel | Perceived Mass: -50%`	Strap tall furniture (Grandfather Clocks, Arcade Cabinets) onto the frame to roll them smoothly up stairs and angled chutes.

Category F: Acoustic, Diagnostic & Remote Gadgets

ID	ITEM NAME	COST	PHYSICAL PARAMETERS	SYSTEMIC GAMEPLAY & PHYSICS USES
F1	**Wind-Up Kitchen Timer**	$90	`0–60s Dial | Ring: 95 dB`	Twist dial, tape to a rolling chair or pendulum rope, and let it ring across the sector to bait *The Tuning Fork* and *The Ram* into each other.
F2	**Digital Hanging Scale**	$130	`Max: 500kg | Exact kg & $`	Clip between rope and loot. Displays exact mass for counterweight math and **instantly exposes disguised Mimics (always 95.0kg)!**
F3	**Walkie-Talkie Pair**	$160	`Relays Voice & Physics Audio`	Talk across deep shafts—OR **leave Walkie B under a ceiling trap in a locked room and yell into Walkie A** from 100m away to trigger an ambush!
F4	**Compressed Air-Horn**	$100	`135 dB Blast | 6s Stun`	Point-blank blast into *The Tuning Fork* overloads its acoustic receptors and ragdoll-stuns it for 6s (while alerting the whole floor).
F5	**Contact Stethoscope**	$170	`10x Wall/Prop Audio Gain`	Press against doors or suspect armchairs to hear monsters in the next room or detect a **sleeping Mimic's heartbeat**.
F6	**RC Motorized Chassis**	$420	`Speed: 6 m/s | Load: 25kg`	Remote-controlled flatbed. Duct-tape a flashlight, timer, or flare onto it to scout vents and herd entities away from your hoist tube.

3. COMPLETE STEALABLE FURNITURE & ANOMALOUS LOOT CATALOG

TIER	FURNITURE / LOOT ITEM	DEPTH	VALUE	MASS	FRAGILITY	SPECIAL TRANSPORT & PHYSICS QUIRK
T1	**Swivel Task Chair**	-20m+	**$90**	8 kg	Low	Built-in caster wheels; great starter door wedge or rolling base for taping boxes on top.
T1	**Brass Bankers Lamp**	-20m+	**$180**	3 kg	High	Glass shade shatters on hard drops (-$120); emits warm light if taped to a battery.
T1	**Potted Monstera Plant**	-30m+	**$240**	22 kg	Medium	Ceramic pot cracks if tipped; spilled soil creates a high-friction patch on slippery floors.
T1	**4-Drawer Filing Cabinet**	-40m+	**$350**	65 kg	Very Low	**Drawers slide out if tilted!** Must tape drawers shut before hoisting or center-of-mass shifts mid-shaft.
T2	**Executive Mahogany Desk**	-80m+	**$750**	90 kg	Low	Too wide (1.8m) for standard single doors upright—must physically rotate 90° onto its side in VR to fit through!
T2	**Mid-Century Velvet Sofa**	-100m+	**$1,400**	115 kg	Medium	Thick cushions dampen collision sound (-50% dB) and act as a soft crash pad at the bottom of a shaft.
T2	**1982 CRT TV Cart**	-120m+	**$1,100**	55 kg	High	Top-heavy rolling cart; smashing the CRT tube triggers a 110 dB implosion pop and electrical spark.
T2	**Antique Grandfather Clock**	-150m+	**$2,200**	95 kg	High	**Ticks constantly (40 dB) and chimes at 105 dB if tilted past 25°!** Must open case and tape the pendulum still before hoisting!
T3	**Venetian Crystal Chandelier**	-200m+	**$3,500**	70 kg	**Extreme**	Hangs 8m up in atrium ceilings. Must grapple up, attach a pulley rope *before* cutting its chain, and wrap in Bubble-Wrap.
T3	**Polybius Arcade Cabinet**	-240m+	**$2,900**	135 kg	High	Randomly plays loud 8-bit attract music (85 dB) when jostled unless Acoustic Foam Tape is placed over its speaker grille.
T3	**Carrara Marble Statue**	-300m+	**$4,800**	220 kg	Medium	Downward weight (2,156N) snaps Hemp Twine and Dynamic Rope—**requires Braided Steel Cable or a 4x Compound Pulley!**
T3	**Gilded Baroque Mirror**	-320m+	**$3,200**	45 kg	**Extreme**	Reflects flashlight beams around corners to freeze ceiling *Counterweights* while keeping your hands free.
T4	**"Helium-Glass" Table**	-400m+	**$6,500**	**-50 kg**	High	**Negative mass—falls UPWARD into the ceiling!** Must tie or tape at least 60kg of ballast weight to keep it on the floor!
T4	**Gyroscopic Armillary**	-450m+	**$7,200**	80 kg	Medium	Spinning rings generate gyroscopic precession torque, fighting turns and swinging wildly on single-rope hoists.
T4	**Superconducting Server**	-500m+	**$9,000**	160 kg	Low	Pulses every 20s, magnetically yanking nearby metal props and tools toward itself unless insulated with Rubber Tape.

TIER	FURNITURE / LOOT ITEM	DEPTH	VALUE	MASS	FRAGILITY	SPECIAL TRANSPORT & PHYSICS QUIRK
★	**Taped-Up Live Mimic**	Any	**$2,500**	95.0 kg	N/A	A sleeping Mimic wrapped in 4+ strips of Kevlar Tape. Thrashes every 12s on the hoist line—sell alive on the Cargo Scale!

4. COMPLETE ENTITY FIELD GUIDE (5 CORE ENTITIES + 3 APEX BOSSES)

ENTITY NAME	PHYSICS STATS	QUIRK & THREAT TO YOUR HEIST	ITEM & PHYSICS COUNTER-STRATEGIES
1. The Ram *(Inertia Anomaly)*	`Mass: 180 kg` `Force: 4,500N`	High linear momentum, awful braking/turning. Charges players and **smashes expensive furniture into $0 splinters**.	• **Tripwires:** String rope 25cm high to flip its active ragdoll at 30 mph. • **Free Hoist Trick:** Tie a door to your heavy loot at the bottom of a shaft—trick The Ram into charging out a window so its momentum **hoists your sofa up the shaft!**
2. The Seep *(Gap Crawler)*	`Mass: 25 kg` `Pull: 350N`	Slithers through 15cm door gaps, vents, and **inside vertical rappel tubes**; chews through Hemp Twine under load.	• **Duct Tape:** Tape vent grates and under-door gaps shut. • **Ball-and-Chain:** Snag it with a Grapple/Rope and tie it to any 50kg+ desk—it lacks the force (350N) to drag the anchor! • **Halogen Light:** High beams burn it back into vents.
3. The Mimic *(Mass Impostor)*	`Mass: 95.0 kg` `Bite: 1,800N`	Disguises as high-dollar furniture. Wakes up and attacks when grabbed with bare hands or loaded into the lift.	• **UV Blacklight / Scale:** Glows with veins under UV; always reads 95.0kg on the Hanging Scale. • **Kevlar Tape Capture:** Wrap 4+ strips of Kevlar Tape around it *while asleep* to weld its jaws shut and **sell it for $2,500!**
4. Counterweight *(Shaft Predator)*	`Ceiling Fixed` `Lift: 1,600N`	Lurks at the top of rappel tubes & ceiling holes; drops sticky cables to hoist players or **steal loot off your rope**.	• **Halogen Floodlight:** Shining light into its hole forces it to drop its catch. • **Free Winch Exploit:** Let it grab your 120kg sofa at the bottom of a shaft, let it reel the sofa to the top floor for you, then blind it so it drops the sofa at the top!
5. Tuning Fork *(Impulse Hunter)*	`Mass: 110 kg` `100% Blind`	Sprints at 12 m/s to the exact 3D coordinate of any collision impulse, rope snap, clock chime, or mic noise.	• **Acoustic Tape + Caster Wheels:** Silent-roll heavy loot right past it. • **Walkie-Talkie / Timer:** Trigger remote acoustic lures two rooms away. • **Air-Horn:** Point-blank 135 dB blast stuns it for 6 seconds.

5. VERTICAL MAP ARCHITECTURE & TECHNICAL EXECUTION PLAN

DEPTH ZONE	SECTOR BIOME	VERTICAL RAPPEL TUBE GEOMETRY	SIGNATURE PHYSICS & EXTRACTION MECHANICS
0m (Surface)	**Cargo Van & Drop Hatch**	3m × 3m breach hatch with overhead winch frame.	Cargo Scale payout zone, Shop Terminal, physical equipment wall racks.
-20m to -100m	**1. The Yellow Annex**	Straight vertical concrete tubes (15m–30m drops).	Hinged lever doors, pop-out drop-ceiling tiles (overhead rope routing), swivel chairs.
-100m to -250m	**2. Brutalist Archive**	Offset elevator shafts requiring pendulum rope swings.	Unpowered counterweight cages, rolling ladders, 8m high crystal chandeliers.

DEPTH ZONE	SECTOR BIOME	VERTICAL RAPPEL TUBE GEOMETRY	SIGNATURE PHYSICS & EXTRACTION MECHANICS
-250m to -400m	**3. Deep Poolrooms**	Ceramic drainpipes & flooded vertical sumps.	**Archimedes Buoyancy** (wood floats up shafts; marble sinks), wet tile (friction 0.02), *The Undertow* boss.
-400m to -600m+	**4. Tilted Hotel & HVAC**	Rotating gimbal shafts & high-wind turbine chutes.	**Shifting Gravity Vectors** (rooms tilt 45° every 3 min) & aerodynamic updrafts that float sail-rigged loot.

Apex Shaft Boss Encounters (Deep Sector Mega-Puzzles)

APEX BOSS	SECTOR & DEPTH	MECHANICAL THREAT	SYSTEMIC PHYSICS SOLUTION
Apex 1: The Pendulum	Brutalist Archive `-200m Shaft`	Colossal concrete-and-brass metronome swinging across the 6-story central hoist shaft, smashing hauled loot.	Harpoon its mass at peak swing ($v = 0$) with Braided Steel Cable and anchor to cracked load-bearing columns so its downward swing collapses the roof onto itself.
Apex 2: The Undertow	Deep Poolrooms `-350m Sump`	Submerged vortex entity in flooded shafts that drags floating rafts and sinking furniture into the drain abyss.	Duct-tape 3 **Inflatable CO2 Lift Bags** to a heavy steel cabinet, let The Undertow snag it, and pull the trigger-rope—rocketing it upward with 3,000N of buoyancy into the ceiling grid.
Apex 3: The Centrifuge	HVAC Chutes `-500m Turbine`	Cyclonic turbine entity alternating between violent suction and violent updraft blasts in the main air shaft.	Use the updraft phase with a taped plywood sail to **float heavy loot up the 200ft shaft for free**, or launch a caster-wheeled steel locker into its core during suction to jam the rotors.

6. UNITY 6 & QUEST 3 PROCEDURAL C# ARCHITECTURE

CORE C# MODULE	PRIMARY RESPONSIBILITY	KEY PHYSICS & PERFORMANCE IMPLEMENTATION DETAILS
`GameBootstrapper.cs`	1-Click Scene & Run Initialization	Generates Drop Hatch, procedural vertical tubes, modular liminal rooms, furniture loot, and player rig at runtime with zero manual Unity scene dependencies.
`DualPhysicsPlayerRig.cs`	OpenXR VR + Desktop Sim Rig	Unified physics hands supporting both OpenXR VR controllers and Mouse/Keyboard Desktop Simulator mode for rapid flat-screen + headset testing.
`TetherAndRappelSystem.cs`	Ropes, Pulleys, Grapples & Rappel	Contextual anchor snapping, `ConfigurableJoint` tension/break math, 2x/4x `PulleyNode` force division, Waist Carabiner friction sliding, and tripwires.
`DuctTapeAndGadgetSystem.cs`	Welding, Wheels, Lights & Tapes	Compound `Rigidbody` merging, strip tear-force stacking, Caster Wheel attachment, Bubble-Wrap dent protection, Acoustic Tape muffling, and UV/Halogen beams.
`LootAppraisalManager.cs`	Furniture $, Denting & Quotas	Tracks `BaseDollarValue`, collision impulse dent penalties, Digital Scale readouts, Cargo Scale payouts, and Shop Terminal upgrades.
`EntityEcosystemAI.cs`	5 Roaming Entities + Infighting	Procedural multi-joint active-ragdoll controllers for Ram, Seep, Mimic, Counterweight, and Tuning Fork, plus room-based physics sleep culling (cap: 30 awake bodies).